KLAi

The chain rule and the Jacobian: how a slope travels backwards through a model made of many steps

A model is a long chain of steps from the input to the loss. How does the slope of the loss reach a number buried many steps deep?

Mathematics for AI and Machine Learning | Session 9 of 18 | Calculus and optimisation
KLAi Professional Learning | Jotham Martin Wambi

Today in one view: what you will leave with

By the end of today you can



Differentiate a function of a function with the chain rule and check the answer with a finite difference

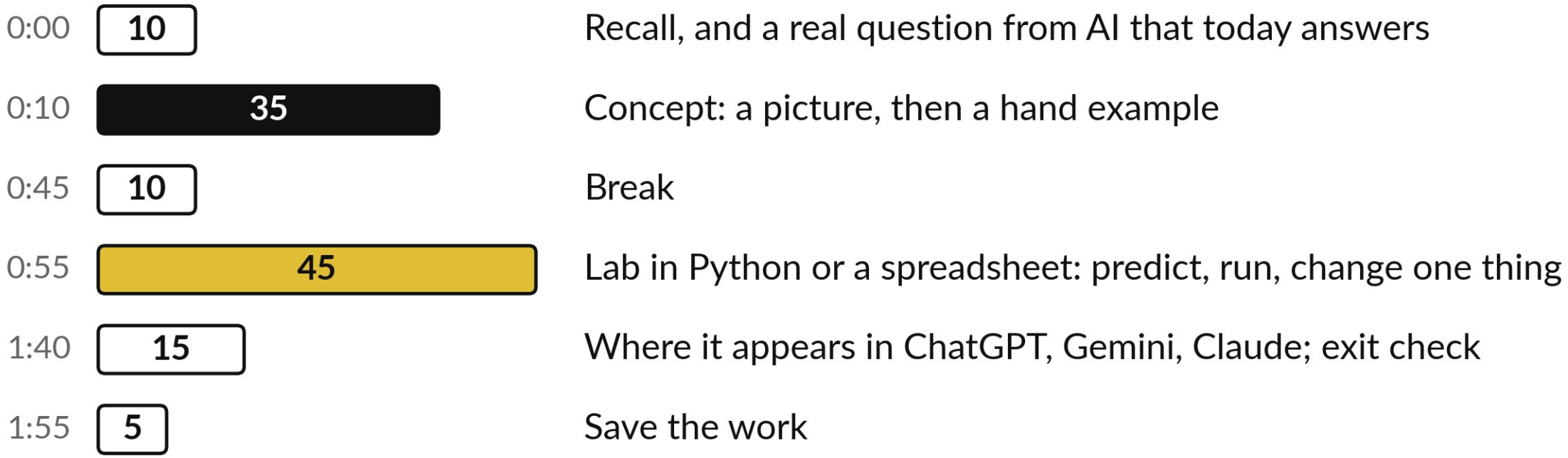


Draw a computation graph, compute forward values and backward derivatives, and add the slopes when a value feeds two paths



Write down and compute the Jacobian matrix of a vector function with two inputs and two outputs

Your two hours: chain rule, graphs, the Jacobian, then the lab



Pictures and hand examples first, then the lab on the same numbers in Python or a spreadsheet.

Three questions from last time

1. What is a partial derivative?

2. What is the gradient, and which way does it point?

3. The loss of the line y-hat = b + w x at w = 1, b = 0 has gradient (−18.5, −6.5). What does it tell you to do?

Answer aloud first. Then we check together.

Answers to the three questions from Session 8

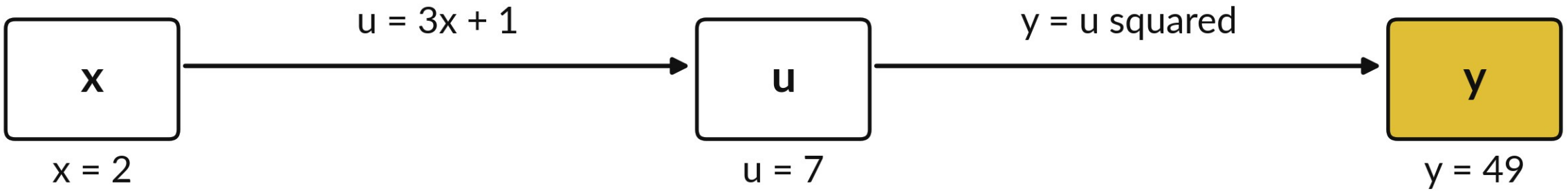
Question	Short answer
What is a partial derivative?	The slope of a function when you change one input and hold all the other inputs fixed.
What is the gradient, and which way does it point?	The column of all the partial derivatives. It points in the direction of steepest increase, so training steps along minus the gradient.
The loss of the line y-hat = b + w x at w = 1, b = 0 has gradient (-18.5, -6.5). What does it tell you to do?	Raise both w and b, because both entries are negative. The w entry is larger in size (18.5 against 6.5), so the loss is more sensitive to w here.

If one answer surprised you, look back at Session 8 before the lab.

Words we use today

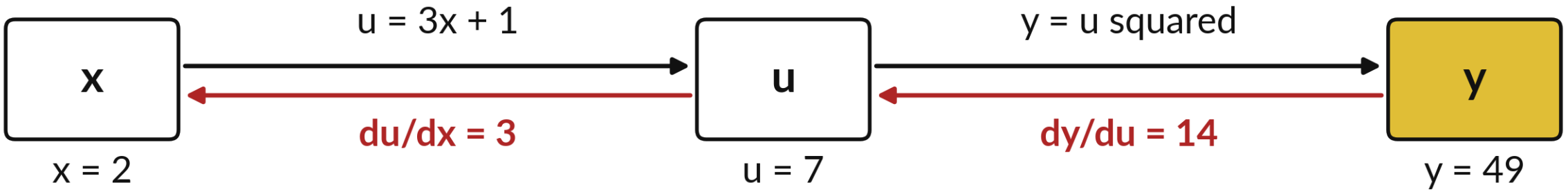
	Word	What it means
	**Composite function**	A function of a function: the output of one function becomes the input of the next. Written f(g(x)).
	**Chain rule**	The derivative of a composite function is the derivative of the outer function times the derivative of the inner function.
	**Computation graph**	A picture of a calculation as boxes (values) and arrows (operations), from the inputs to the loss.
	**Jacobian**	The matrix of all the partial derivatives of a function with several inputs and several outputs. One row for each output, one column for each input.
	**Backpropagation**	Computing the derivatives of the loss by applying the chain rule backwards through the computation graph.

A function of a function: the output of one step becomes the input of the next



Two steps. The input x = 2 becomes u = 7, and u becomes y = 49.

The chain rule: slopes along the chain multiply



A tiny step in x moves u 3 times as far; in u, y 14 times as far. So y moves 3 × 14 = 42 times as far.

The chain rule: dy/dx is dy/du times du/dx

Chain rule

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Our data at x = 2

$$\frac{dy}{dx} = 14 \times 3 = 42$$

Symbol	What it means	In our data
y = f(u)	The outer function: it takes u and returns y.	y = u squared; dy/du = 2u = 14 at u = 7
u = g(x)	The inner function: it takes x and returns u.	u = 3x + 1; du/dx = 3
dy/dx	The slope of the whole chain: how fast y changes when x changes.	42 at x = 2

Evaluate dy/du at the value of u that the forward step produced (u = 7), never at x.

Check the chain rule with a finite difference: y = (3x ✦ 1) squared at x = 2

Step h	y(2 ✦ h)	y(2)	Forward slope = (y(2 ✦ h) minus y(2)) / h
1	100	49	51.0000
0.1	53.29	49	42.9000
0.01	49.4209	49	42.0900
0.001	49.042009	49	42.0090

The slopes settle on 42, the chain rule answer. The central difference with h = 0.001 gives 42.0000. In the lab you check every chain rule answer this way.

The chain rule on three functions at x = 1

Function y(x)	Inner u	Outer in u	du/dx	dy/du	dy/dx at x = 1
y = (3x ✦ 1) squared	3x + 1	u squared	3	2u = 8	24
y = e to the 2x	2x	e to the u	2	e^u = 7.389	14.778
y = ln(x squared ✦ 1)	x squared + 1	ln u	2x = 2	1/u = 0.5	1

The second row explains the remark in Session 7: e to the 2x has slope 2 times e to the 2x.

Keep the inner slope, and multiply the slopes along a chain

Forgetting the inner slope

The slope of u squared is 2u.

But u = 3x + 1 also changes with x.

Multiply by du/dx = 3 as well.

Adding along a chain

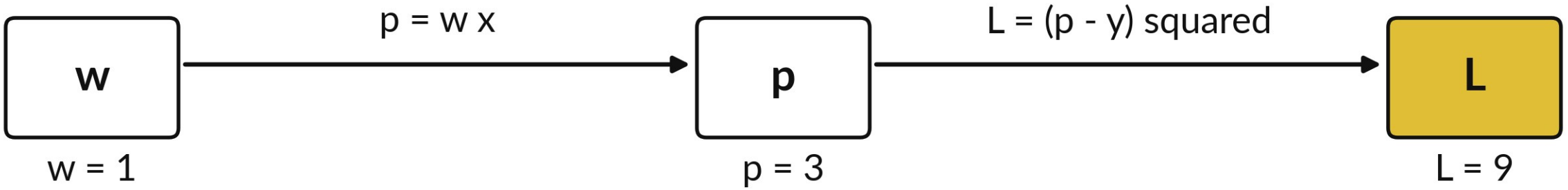
Slopes along a chain multiply.

14 + 3 = 17 is wrong here.

14 times 3 = 42 is right.

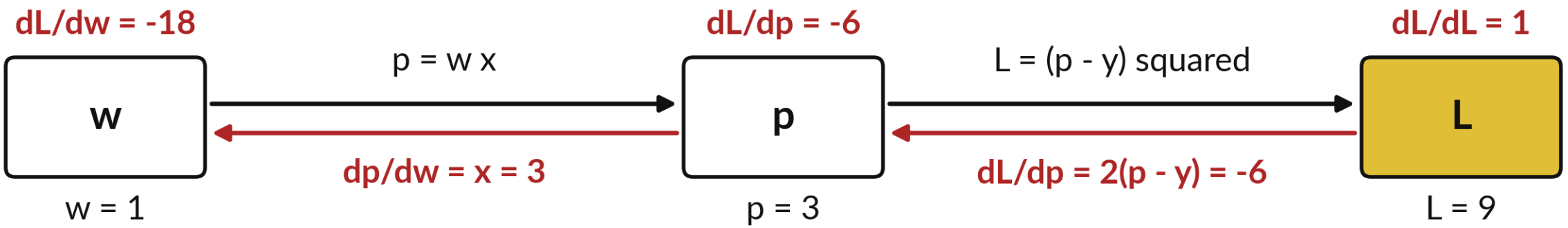
Outer slope times inner slope, each evaluated at its own value.

A computation graph draws a calculation as boxes for values and arrows for operations



One delivery, x = 3 km, y = 6. Forward pass: p = 1 × 3 = 3, then L = (3 - 6) squared = 9.

Backward: each box gets the slope of the loss with respect to it



Black: forward values. Red: the backward pass. Start at 1 and multiply by each local slope.

Backward pass on the tiny graph: multiply the local slopes along the path

Chain rule on the graph

$$\frac{dL}{dw} = \frac{dL}{dp} \cdot \frac{dp}{dw}$$

Our data

$$\frac{dL}{dw} = (-6) \times 3 = -18$$

Symbol	What it means	In our data
dL/dp	Local slope of the loss in p: 2 (p - y). The 2 of Sessions 7 and 8.	−6
dp/dw	Local slope of p in w: p = w x, so x. The x_i of Sessions 7 and 8.	3
dL/dw	The slope of the loss in w: the product of the two.	−18

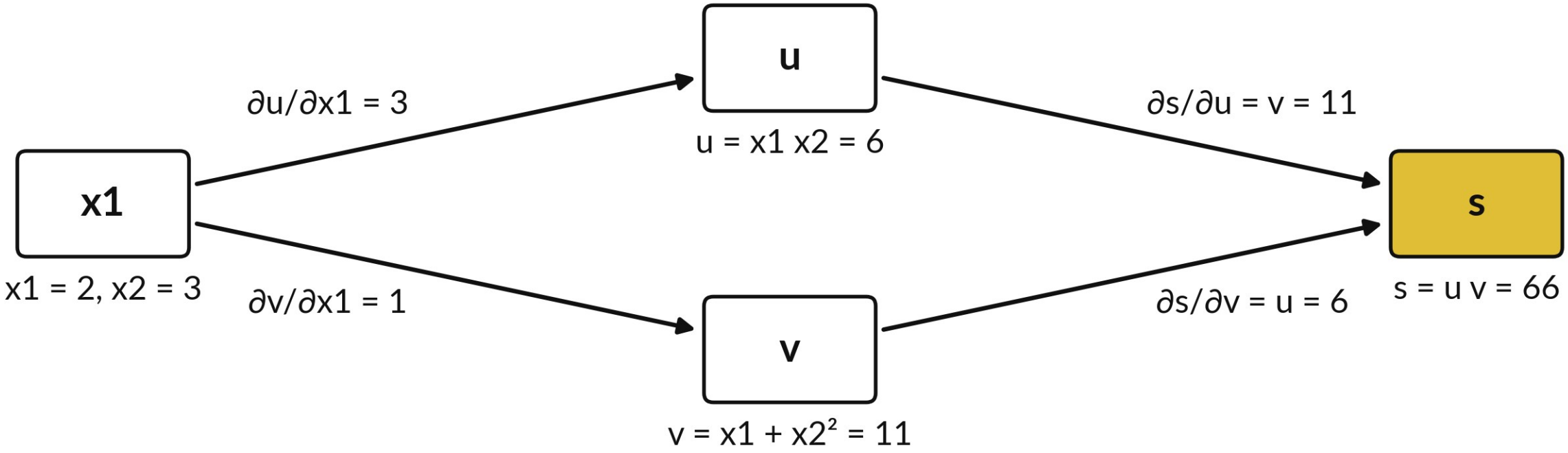
A local slope is the derivative of one box's output with respect to its own input.

Backward values for all four deliveries at w = 1: the average is the Session 8 slope

Row i	Distance x	Error p minus y	dL/dp = 2 e	dL/dw = x times dL/dp
1	1	−2	−4	−4
2	2	−3	−6	−12
3	3	−3	−6	−18
4	4	−5	−10	−40
Mean				−18.5

Row 3 is the graph we just drew: −6 and −18. The mean loss has the mean of the four backward values: −18.5, as in Sessions 7 and 8.

When one input feeds two boxes, the slopes along the two paths add



Through u: 11 × 3 = 33. Through v: 6 × 1 = 6. Add the paths: ∂s/∂x1 = 39.

The multivariable chain rule: multiply along each path, then add the paths

Slope of s in x1

$$\frac{\partial s}{\partial x_1} = \frac{\partial s}{\partial u}\frac{\partial u}{\partial x_1} + \frac{\partial s}{\partial v}\frac{\partial v}{\partial x_1} = 11 \cdot 3 + 6 \cdot 1 = 39$$

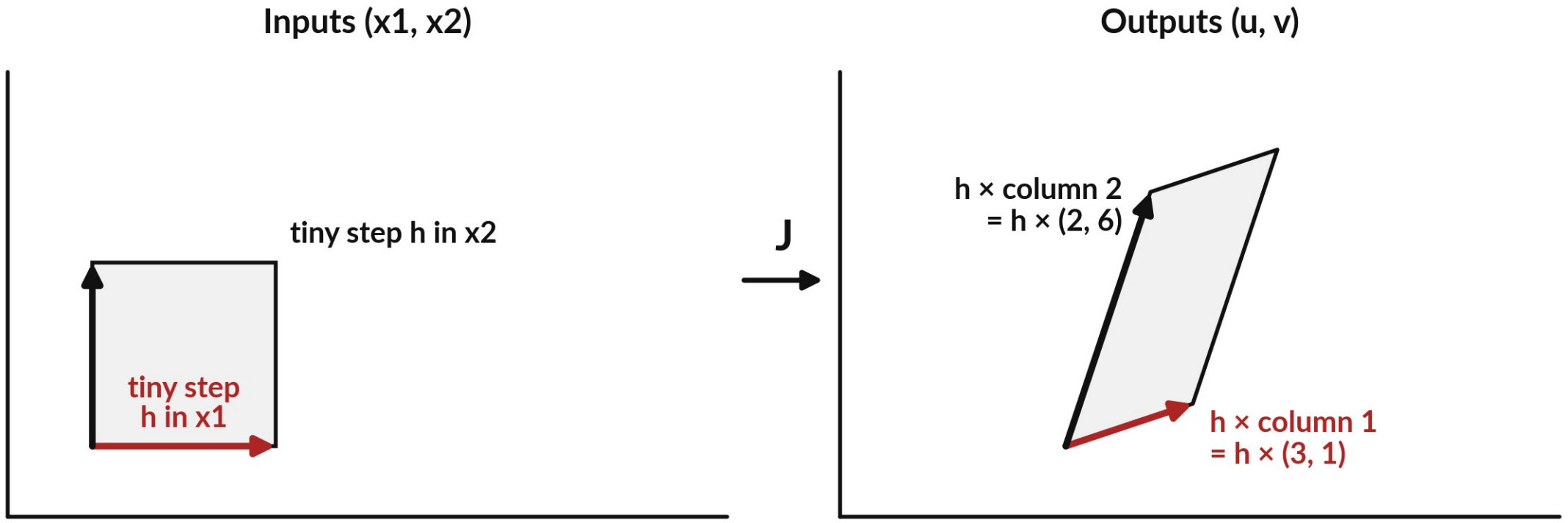
Slope of s in x2

$$\frac{\partial s}{\partial x_2} = \frac{\partial s}{\partial u}\frac{\partial u}{\partial x_2} + \frac{\partial s}{\partial v}\frac{\partial v}{\partial x_2} = 11 \cdot 2 + 6 \cdot 6 = 58$$

Symbol	What it means	In our data
u, v	Two boxes fed by the inputs: u = x1 x2 and v = x1 + x2 squared.	u = 6, v = 11 at (2, 3)
∂s/∂u, ∂s/∂v	The slopes of s = u v in each box: v and u.	11 and 6
∂u/∂x2, ∂v/∂x2	Local slopes of the boxes in x2: x1 and 2 x2.	2 and 6

Check: s = $x1^2$ x2 + x1 $x2^3$, so ∂s/∂x1 = 2 x1 x2 + $x2^3$ = 12 + 27 = 39 and ∂s/∂x2 = $x1^2$ + 3 x1 $x2^2$ = 4 + 54 = 58.

A vector function has several outputs: the Jacobian holds every slope



A tiny square with sides h becomes almost a parallelogram: its sides are h times the columns of J.

The Jacobian: one row for each output, one column for each input

Jacobian entry

$$J_{ij} = \frac{\partial F_i}{\partial x_j}$$

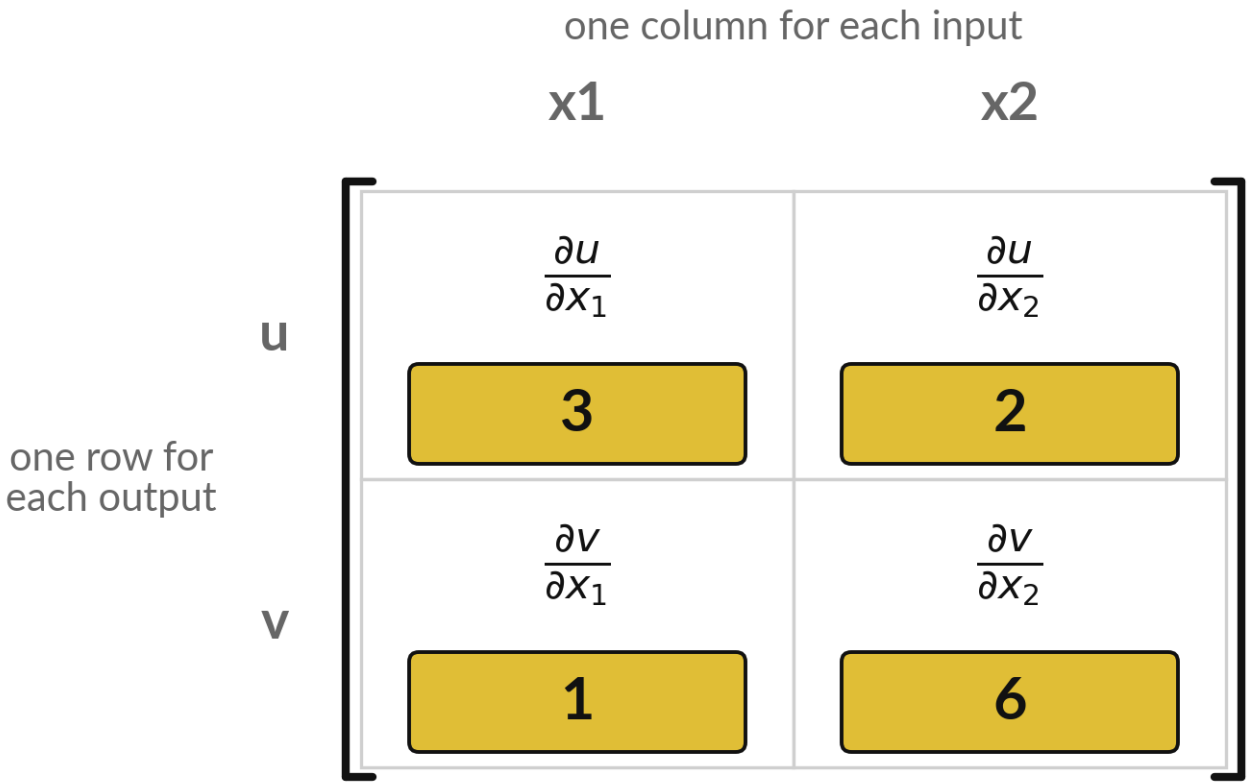
Our function

$$u = x_1 x_2 \qquad v = x_1 + x_2^2$$

Symbol	What it means	In our data
F_i	Output i of the function F. Here F_1 = u, F_2 = v.	at (2, 3): u = 6, v = 11
x_j	Input number j. Here x_1 and x_2.	the point (2, 3)
J_ij	The slope of output i in input j. J has 2 rows and 2 columns.	J = rows (3, 2) and (1, 6)

Row 1: ∂u/∂x1 = x2 = 3, ∂u/∂x2 = x1 = 2. Row 2: ∂v/∂x1 = 1, ∂v/∂x2 = 2 x2 = 6.

The Jacobian drawn as a matrix, with the numbers at the point (2, 3)



Each gold number is one partial derivative at the point (2, 3).

Row u: x2 = 3 and x1 = 2. Row v: 1 and 2 x2 = 6.

Check: nudging x2 from 3 to 3.001 raises v by 0.006001, so 6.001 per unit.

One nudge of one input gives a whole column.

Stack the two path sums and you get J transposed times the slopes of the outputs

Gradient of s in the inputs

$$\nabla_x s = \begin{pmatrix} \partial s/\partial x_1 \\ \partial s/\partial x_2 \end{pmatrix} = \begin{pmatrix} 3 \cdot 11 + 1 \cdot 6 \\ 2 \cdot 11 + 6 \cdot 6 \end{pmatrix} = \begin{pmatrix} 39 \\ 58 \end{pmatrix}$$

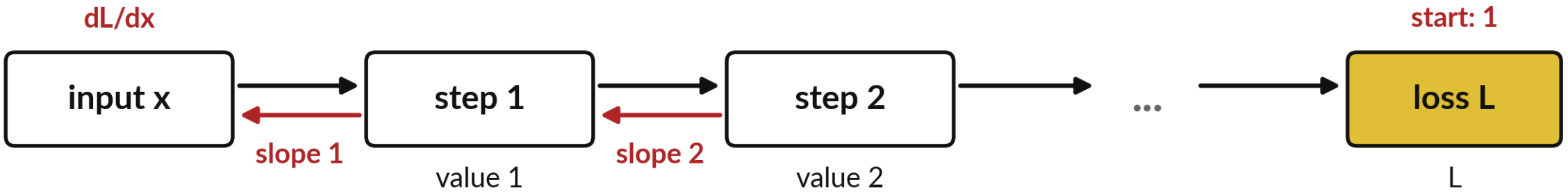
The same, as a matrix

$$\nabla_x s = J^T \nabla_{(u,v)} s, \qquad \nabla_{(u,v)} s = \begin{pmatrix} \partial s/\partial u \\ \partial s/\partial v \end{pmatrix} = \begin{pmatrix} 11 \\ 6 \end{pmatrix}$$

Symbol	What it means	In our data
nabla_x s	The gradient of s with respect to the inputs x1 and x2: a column, as in Session 8.	(39, 58)
nabla_(u,v) s	The gradient of s with respect to the outputs u and v: s = u v, so (v, u).	(11, 6)
J transposed	Rows and columns of J swapped. Row 1 of J^T is column 1 of J, (3, 1): the slopes of u and v in x1.	rows (3, 1) and (2, 6)

Each row of J^T dotted with (11, 6) is one path sum. In text, columns are written across, as (11, 6).

Many steps: the backward value is the product of the local slopes along the whole chain



Start at the loss with 1. Going back, multiply by each local slope (red). Where paths split, add them.

Four lines of Python run the tiny graph forward and then backward

```
p = w * x
L = (p - y) ** 2
dL_dp = 2 * (p - y)
dL_dw = dL_dp * x
print(p, L, dL_dp, dL_dw)
```

```
3.0 9.0 -6.0 -18.0
```

In plain words

Lines 1 and 2: the forward pass. p is the prediction, L the loss.

Line 3: the local slope of the loss in p, 2 (p - y).

Line 4: times the local slope of p in w, which is x. This is the chain rule.

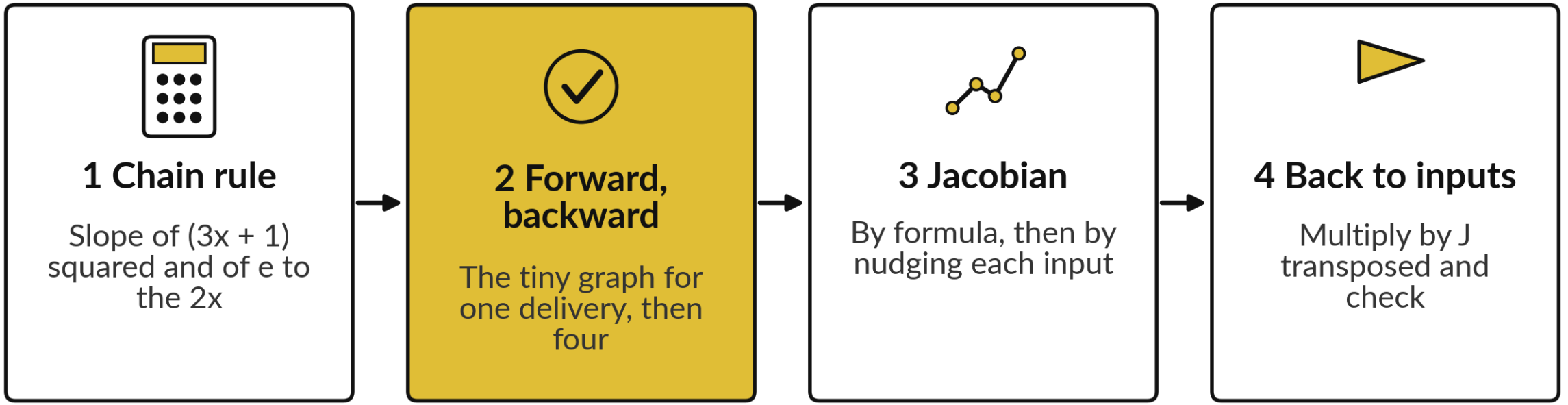
With w = 1, x = 3, y = 6 it prints 3, 9, −6 and −18.

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Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab checks every slope today twice: once by rule and once by nudging



Python: Steps 1 to 6 of the notebook. Spreadsheet: sheets Chain, Backward and Jacobian. Predict first, then run.

Expect 42, then -6 and -18, then a Jacobian with rows (3, 2) and (1, 6)

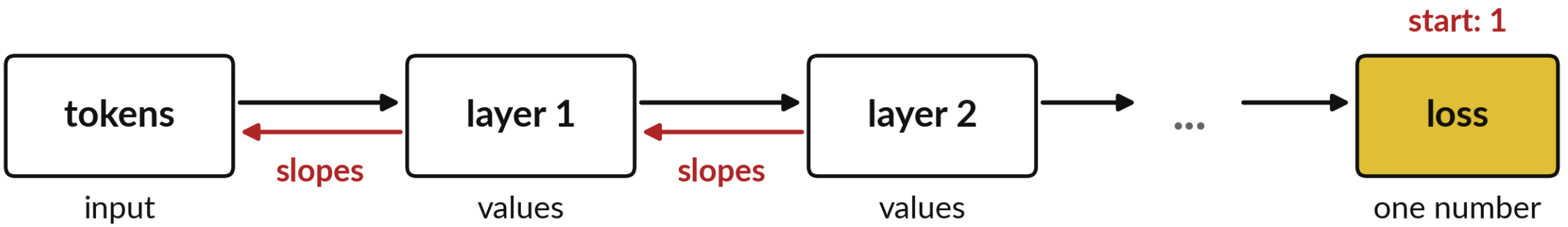
	What you compute	You should see
	Chain rule slope of (3x ✦ 1) squared at x = 2	42
	Chain rule slope of e to the 2x at x = 1	14.778
	Forward values for the delivery x = 3, y = 6 at w = 1: p and L	3 and 9
	Backward values: dL/dp and dL/dw	−6 and −18
	Mean of the four dL/dw values at w = 1	−18.5
	Jacobian of (x1 x2, x1 ✦ x2 squared) at (2, 3)	rows (3, 2) and (1, 6)
	Gradient of s = u v in the inputs, J transposed times (11, 6)	(39, 58)

If your number differs, stop and ask. Do not change a number to match. Find the step that went wrong.

A slope of 14 or a flipped Jacobian each have a simple cause

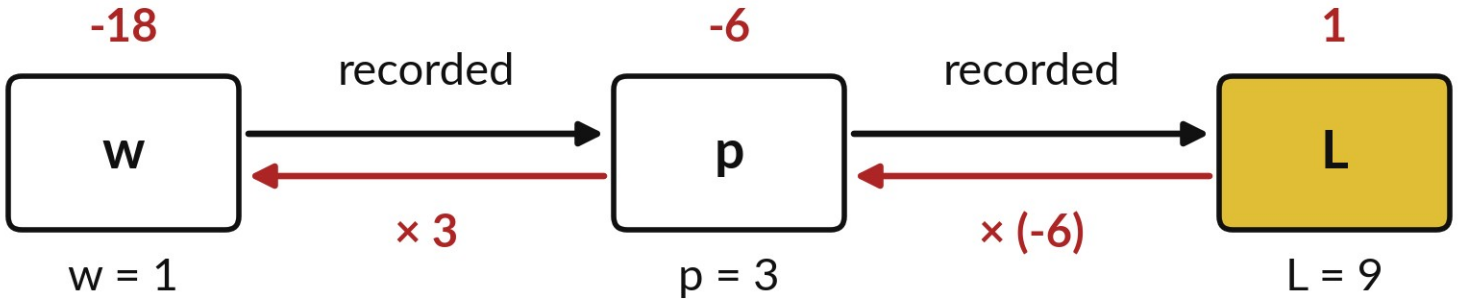
	What you see	What it means	What to do
	**NameError in Python**	A cell that defines a name was skipped	Run all cells from the top, in order
	**Chain slope is 14 instead of 42**	The inner slope du/dx was left out	Multiply by 3 as well as by 2u
	**Jacobian looks flipped**	Rows and columns swapped: outputs are rows	Row i is output i, column j is input j
	**Spreadsheet shows #VALUE!**	A range with text or a gap in a formula	Check that every cell in the range holds a number

Where this appears in AI: the chain rule runs backwards through every layer, and every step inside it



Red: backward slopes. Meta's Llama 3 paper (2024) lists 32 and 80 transformer layers (8B, 70B).

Where this appears in AI: libraries record the graph on the way forward and replay it backwards



PyTorch (loss.backward()) and JAX (jax.grad) record each step of the forward pass, then run the chain rule backwards.

This is automatic differentiation. Nudge checks like today's are how it is tested.

Libraries never build a layer's full Jacobian. They multiply by J transposed directly: a vector-Jacobian product.

Ask an AI assistant for this Jacobian and check which entry is which

8 minutes **Hands-on, with any assistant**

Ask an assistant for the Jacobian of F(x1, x2) = (x1 x2, x1 ✦ x2 squared) at the point (2, 3).

Ask it to explain each entry.

Compare with ours: rows (3, 2) and (1, 6).

Run your lab Jacobian code.

If it differs, find the first entry that differs.

You keep

One line: its Jacobian, yours, match or not. Never paste confidential data into an assistant.

Exit check: a chain of slopes and the shape of a Jacobian

Question	Short answer
What is the slope of y = (2x ✦ 5) cubed at x = 0?	150. Outer: 3u squared = 75. Inner: 2. Product 150.
A chain has three steps with local slopes 2, -3 and 0.5. What is the slope of the output with respect to the input?	2 times −3 times 0.5 = −3.
What is the shape of the Jacobian of F(x1, x2) = (2 x1, x1 ✦ x2), and what are its entries?	2 rows (outputs) by 2 columns (inputs): rows (2, 0) and (1, 1).

Write your own answer before you read ours. The automatic check counts the lab numbers.

Save the lab and keep the chain rule slope, the backward value and the Jacobian

What you keep from today

Keep your saved lab. Write down the chain rule slope 42, the backward value −18 for the tiny graph, and the Jacobian, rows (3, 2) and (1, 6).

Our question today

A model is a long chain of steps from the input to the loss. How does the slope of the loss reach a number buried many steps deep?

Next session

Session 10: gradient descent.

Take away

Saved lab
Notebook or workbook

Chain rule slope
42

Backward value
dL/dw = -18

Next session
Gradient descent